

Phase 3.2 - v32.17 Working Document

Perturbative Response Law for Local Hotspot Formation

Research plan, mathematical target, numerical protocol, and decision gates

1 Purpose of v32.17

Phase 3.2 has reached a point where further system-size scaling is no longer the highest-value next step. The 7Q–12Q robustness study has established that local hotspot prominence is repeatedly associated with basis-invariant, degeneracy-closed interference and coherence in the effective P -space feedback generated through virtual Q -space pathways.

The purpose of **v32.17** is therefore not to add another correlation feature. It is to test whether hotspot prominence can be connected more directly to the *response* of the effective operator Σ_E under a controlled perturbation.

The central question is:

$$\boxed{\text{Is hotspot prominence governed by the local perturbative sensitivity of } \Sigma_E?} \quad (1)$$

If the answer is positive, Phase 3.2 moves from a static structural correlation toward a candidate response law derived from the operator mechanism itself.

2 Frozen background from v32.14–v32.16.1

The relevant Hilbert-space decomposition is built from a complete degeneracy-closed target subspace P and its complement $Q = I - P$. For a target energy E , the second-order effective feedback is written as

$$\Sigma_E = \sum_G T_G, \quad (2)$$

with degeneracy-closed channel contributions

$$T_G = g_\eta(E - E_G) P V P_G V P, \quad (3)$$

where P_G is the projector onto a complete degenerate eigenspace in Q and g_η is the regularized energy-denominator factor used in the existing Phase 3 implementation.

Two basis-invariant local descriptors have already been identified:

$$A_E^{(G)} = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{(\sum_G \|T_G\|_F)^2}, \quad (4)$$

and

$$C_E^{(G)} = \frac{\|\sum_G T_G\|_F}{\sum_G \|T_G\|_F}. \quad (5)$$

The identity

$$\|\Sigma_E\|_F^2 = \sum_G \|T_G\|_F^2 + 2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K) \quad (6)$$

shows explicitly that the effective norm contains both an incoherent channel-strength term and an interference term.

The 7Q–12Q study showed that $A_E^{(G)}$ and $C_E^{(G)}$ remain positively associated with hotspot prominence over the investigated finite-size range, but neither quantity by itself is a deterministic hotspot law. This motivates a response-based analysis.

3 Core hypothesis for v32.17

The working hypothesis is that a hotspot is characterized not merely by large static interference, but by an unusually strong *change* in the effective virtual-pathway structure under a small perturbation.

Introduce a scalar perturbation parameter λ :

$$H(\lambda) = H_0 + \lambda V. \quad (7)$$

The principal response candidate is

$$R_E = \left\| \frac{d\Sigma_E}{d\lambda} \right\|_F. \quad (8)$$

A second, especially attractive scalar candidate is

$$D_E = \left| \frac{d}{d\lambda} \|\Sigma_E\|_F^2 \right|. \quad (9)$$

The hypothesis to test is therefore

$$\boxed{\mathcal{H}(E) \sim R_E \quad \text{or} \quad \mathcal{H}(E) \sim D_E,} \quad (10)$$

where $\mathcal{H}(E)$ denotes the already-defined hotspot prominence measure.

4 Why the derivative may be more fundamental than the static descriptors

A static value such as $A_E^{(G)}$ can be large while changing very little under the perturbation. Such a point may have strong internal interference without being especially sensitive. Conversely, a moderate static interference pattern may reorganize sharply under a small perturbation and therefore create a strong hotspot response.

This gives a more specific physical picture:

$$\boxed{\text{Hotspot strength may reflect the susceptibility of virtual-path interference.}} \quad (11)$$

The derivative of the squared effective norm can be expanded directly:

$$\frac{d}{d\lambda} \|\Sigma_E\|_F^2 = \frac{d}{d\lambda} \left[\sum_G \|T_G\|_F^2 + 2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K) \right] \quad (12)$$

$$= 2 \sum_G \text{Re Tr} \left(T_G^\dagger \frac{dT_G}{d\lambda} \right) \quad (13)$$

$$+ 2 \sum_{G < K} \text{Re Tr} \left[\left(\frac{dT_G}{d\lambda} \right)^\dagger T_K + T_G^\dagger \frac{dT_K}{d\lambda} \right]. \quad (14)$$

This decomposition separates the response into two interpretable pieces:

- **Channel-strength response:** changes in the individual $\|T_G\|_F$ contributions.
- **Interference response:** changes in the cross terms between different virtual pathways.

If hotspot prominence is mainly driven by the second contribution, v32.17 would give a much sharper mechanistic statement than the static A_G/C_G correlations alone.

5 Two response models to separate mechanisms

The experiment should deliberately distinguish two levels of response.

5.1 Model A: frozen-projector response

In the first model, the reference eigenspaces are kept fixed while the perturbation amplitude changes. Conceptually, P , P_G , and the reference energy labels are frozen. Only the perturbative coupling is varied.

This isolates the response of the already-identified virtual pathways and answers:

How much of the hotspot can be explained without moving the eigenspaces?

(15)

This model is mathematically cleaner and should be implemented first.

5.2 Model B: full spectral response

In the second model, the perturbed Hamiltonian is re-diagonalized at $\lambda = \pm\delta$. The target and Q eigenspaces are then reconstructed consistently, respecting degeneracy closure.

This incorporates eigenspace rotation, energy shifts, and changes in the decomposition itself.

The difference

$$\Delta_{\text{geom}} = R_E^{\text{full}} - R_E^{\text{frozen}} \quad (16)$$

is not necessarily a final observable, but it provides a direct diagnostic of whether the hotspot mechanism depends materially on spectral geometry beyond the frozen virtual-channel structure.

6 Numerical derivative protocol

The first implementation should use a symmetric central difference:

$$\frac{d\Sigma_E}{d\lambda} \approx \frac{\Sigma_E(+\delta) - \Sigma_E(-\delta)}{2\delta}. \quad (17)$$

Correspondingly,

$$R_E(\delta) = \left\| \frac{\Sigma_E(+\delta) - \Sigma_E(-\delta)}{2\delta} \right\|_F, \quad (18)$$

and

$$D_E(\delta) = \left| \frac{\|\Sigma_E(+\delta)\|_F^2 - \|\Sigma_E(-\delta)\|_F^2}{2\delta} \right|. \quad (19)$$

The same construction should be applied to the two existing local descriptors:

$$A'_E(\delta) = \frac{A_E(+\delta) - A_E(-\delta)}{2\delta}, \quad (20)$$

$$C'_E(\delta) = \frac{C_E(+\delta) - C_E(-\delta)}{2\delta}. \quad (21)$$

The derivative step δ must not be trusted at a single value. A small convergence scan should be performed, for example over a short logarithmic sequence around the scale already used for the perturbative experiments. The aim is to identify a stable derivative plateau rather than optimize a correlation.

7 Experimental sequence

The recommended sequence is intentionally conservative.

1. **Start at 8Q, one seed only.** Use the frozen 11-term Hamiltonian setup and one known reference seed. This keeps iteration fast.
2. **Implement frozen-projector response first.** Compute R_E , D_E , A'_E , and C'_E .
3. **Check derivative stability.** Repeat for several δ values and reject any response quantity that changes erratically with step size.
4. **Compare directly with existing hotspot prominence.** Use Spearman rank correlation as the primary first-pass statistic, because the previous Phase 3 analyses have been rank-oriented and do not assume linearity.
5. **Compare against the static baseline.** The new response variables should be judged relative to the existing A_G and C_G performance, not in isolation.
6. **Add the full spectral response only after the frozen version works numerically.**
7. **Expand to all 12 seeds at 8Q only if the pilot is promising.**
8. **Only then generalize across qubit count.** A successful 8Q result can be tested at 7Q–12Q using the already-established finite-size protocol.

8 Primary measurements and comparison table

For each target eigenspace and perturbation family, the pilot should record at least the following quantities:

Quantity	Meaning
$\mathcal{H}(E)$	Existing hotspot prominence; fixed target variable.
$A_E^{(G)}$	Static degeneracy-closed interference descriptor.
$C_E^{(G)}$	Static degeneracy-closed coherence descriptor.
R_E	Frobenius norm of $d\Sigma_E/d\lambda$.
D_E	Absolute derivative of $\ \Sigma_E\ _F^2$.
A'_E	Perturbative derivative of the interference descriptor.
C'_E	Perturbative derivative of the coherence descriptor.
D_E^{self}	Optional contribution from individual channel-strength response.
D_E^{cross}	Optional contribution from changing interference cross terms.

The last two quantities are especially useful if D_E is promising, because they identify which part of the derivative carries the hotspot signal.

9 Decision gates

v32.17 should be hypothesis-driven. The code should not evolve into another broad feature search. The following gates are recommended.

9.1 Gate 1: numerical validity

A candidate derivative is accepted for interpretation only if its ranking and scale are reasonably stable over a small δ range. If the result is dominated by finite-difference noise or level-matching instabilities, it is not a valid mechanism test.

9.2 Gate 2: pilot relevance

At 8Q, one or more response quantities should show a clearly stronger or more systematic association with hotspot prominence than would be expected from a numerically unstable derivative. The exact threshold should not be tuned after seeing the result; the main comparison is qualitative and relative to the frozen A_G/C_G baseline.

9.3 Gate 3: cross-seed confirmation

If the pilot is promising, the response law must retain a positive direction across the 12 frozen seeds. Median Spearman ρ , positive-seed count, and the same sign-test logic already used in the robustness analysis should be reported.

9.4 Gate 4: mechanism decomposition

If D_E or R_E works, determine whether the signal is carried mainly by channel-strength response, interference response, or eigenspace motion. This is the point at which a genuine local law may

become possible.

10 What would count as success?

There are three scientifically useful outcomes.

Outcome A - strong success. A response quantity such as D_E or R_E consistently outperforms the static descriptors and remains robust across seeds. This would justify a candidate hotspot response law of the form

$$\boxed{\mathcal{H}(E) \approx F\left(\frac{d\Sigma_E}{d\lambda}\right)} \quad (22)$$

with the next step being an analytic reduction of F .

Outcome B - partial success. The derivative signal is real but does not replace A_G/C_G . Then the hotspot mechanism is likely two-factor: static pathway organization sets the susceptibility landscape, while the perturbation response determines the observed prominence.

Outcome C - negative result. The derivatives do not improve or stabilize the hotspot relation. This is still valuable: it would tell us that the local law is not a simple first-order susceptibility of Σ_E , and would sharply constrain the next theoretical direction.

11 Important safeguards

Several issues must be controlled explicitly.

- **Degeneracy closure must remain exact.** No return to individual basis vectors inside degenerate eigenspaces.
- **No post-hoc feature fishing.** The quantities listed in this document are the predeclared v32.17 targets.
- **Energy/eigenspace matching must be deterministic.** This is particularly important for the full spectral response.
- **Perturbation scale must be checked, not optimized.** The derivative scan is a numerical convergence test, not a hyperparameter search.
- **REAL and NULL logic must remain separated.** The same control philosophy used previously should be retained if a NULL comparison is introduced.
- **Do not overclaim causality.** Even a strong derivative association would be a mechanistically motivated response law, not by itself a formal proof of causation.

12 Recommended code structure

The new implementation should be a new file, leaving all v32.16.1 code untouched. A suitable name is

`v32_17_perturbative_response.py`

A clean internal structure would contain functions conceptually equivalent to:

- load the frozen model and reference eigenspaces;
- construct degeneracy-closed P and P_G projectors;
- evaluate $\Sigma_E(\lambda)$ and grouped $T_G(\lambda)$;
- compute central-difference derivatives;
- split D_E into self and cross/interference response;
- calculate rank correlations with the frozen hotspot prominence;
- save one-row-per-target CSV output plus a compact seed-level summary.

The first code version should be deliberately small. No cross-qubit loop is needed until the 8Q pilot passes the numerical and conceptual checks.

13 Expected output files

A minimal v32.17 pilot should create:

- `v32_17_seed25042000_rows.csv` - target-level quantities;
- `v32_17_seed25042000_stats.csv` - Spearman correlations and derivative-stability summary;
- `v32_17_delta_scan.csv` - convergence across δ ;
- a short plain-text run summary suitable for direct inclusion in the Phase 3 notes.

If the pilot succeeds, the same schema can be extended to all seeds without changing definitions.

14 Interpretive target

The strongest conceptual statement we are testing is not

$$\text{“hotspots occur where interference is large.”} \quad (23)$$

It is the more specific proposition

hotspots occur where the effective virtual-pathway interference is unusually susceptible to perturbation. (24)
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This would naturally connect the empirical hotspot observable to the operator-level structure already established in Phase 3.2.

15 Stop condition for v32.17

v32.17 should stop after one of two outcomes:

1. a response quantity survives numerical convergence and cross-seed confirmation strongly enough to justify a dedicated analytic derivation; or
2. the response hypothesis fails cleanly, in which case the negative result is documented and the project moves to a different mechanism class.

The project should *not* continue adding ad hoc descriptors within v32.17. The value of this stage is precisely that it tests a narrow, operator-derived hypothesis.

16 Working conclusion

The evidence accumulated through v32.16.1 supports a two-layer picture:

$$\boxed{\text{global symmetry backbone} + \text{local virtual-pathway organization} \longrightarrow \text{hotspot structure}} \quad (25)$$

v32.17 tests the next logical refinement:

$$\boxed{\text{local hotspot prominence} \overset{?}{\longleftarrow} \text{perturbative susceptibility of } \Sigma_E} \quad (26)$$

If successful, this would provide the most direct route so far from the Phase 3 operator formalism to a compact, deduced hotspot law.